FINA2204 Tutorial 1: Introduction

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Jul 2026

Tutorial Information

How is participation calculated?

- Assessment: **10%** of the final grade
- Tutorial: one hour per week, weeks 2 – 12 (study break: week 7)
- Final mark is based on your **<u>best eight tutorial scores</u>**.

$$Participation\ mark = \frac{Best\ 8\ tutorial\ scores}{8} \times 10$$

- For each of the 10 tutorials, you will receive: (i) 0: Absent; (ii) 0.8: Attended but did not actively engage; (iii) 1: Attended and made at least one meaningful contribution.

Tutorial Information

How is participation recorded?

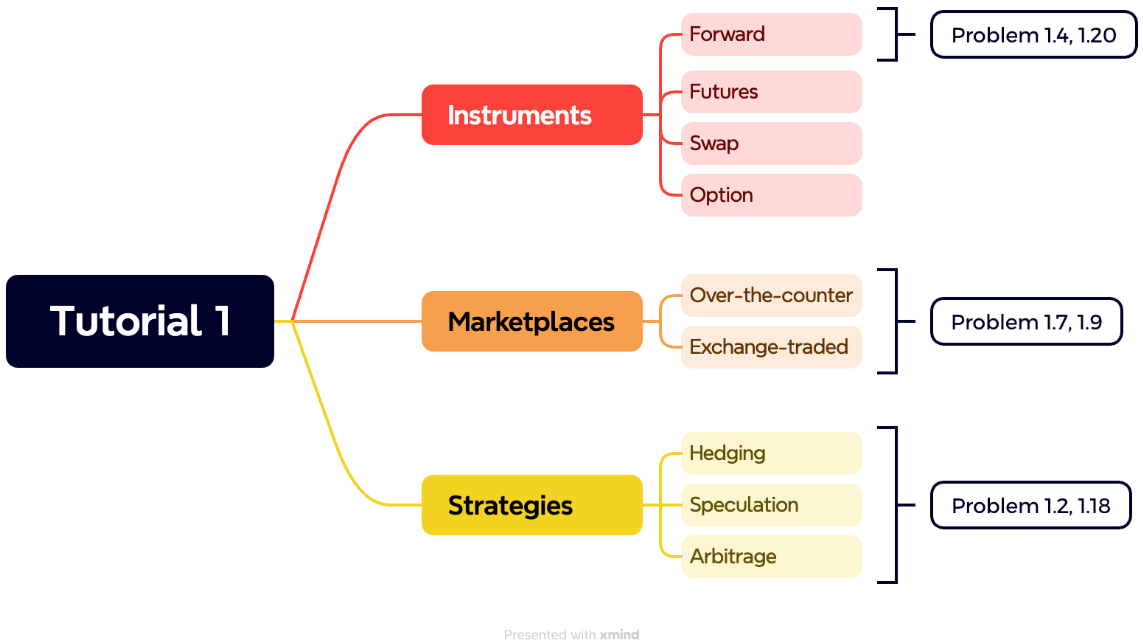
- Attendance sheet: A sign-in sheet will be passed around during each tutorial.
- Attendance: Sign beside your name at the beginning of the class.
- Engagement: At the end of the tutorial, tick the engagement box beside your signature if you contributed at least once. Claims made later **will not** be accepted.
- **Be honest**: Falsely claiming engagement may result in the loss of your participation mark for that tutorial.

Tutorial Information

- All tutorial slides will be shared on my personal **Github page**.
- Link: https://github.com/zrsong/FINA2204_Tutorial_26S2
- The materials will be updated **weekly**, usually by the end of each week.
- Textbook: Hull, J C., *Fundamentals of Futures and Options Markets*, Global Edition, 9th ed., Pearson.
- Email: zirui.song@uwa.edu.au

Agenda

- Problem 1.2
- Problem 1.4
- Problem 1.7
- Problem 1.9
- Problem 1.18
- Problem 1.20



Recap: Derivative

What is derivative?

- A derivative contract is an instrument whose value depends on, or is derived from, the value of another asset (the **underlying**).
- Types of the underlying: (1) shares and equity indices; (2) currencies and interest rates; (3) commodities: gold, oil, wheat, iron ore; (4) Even credit risk and electricity.
- **Instruments** include forwards, futures, swaps, and options.
- The derivate is therefore NOT valued in isolation (e.g., call option on a share; currency forward).

Recap: Derivative

Type	One-line definition	Venue
Forward	Contract to buy or sell an asset at a certain future time for a certain price	OTC
Futures	Same economic idea as a forward, but standardised and traded on an exchange	EXCHANGE
Swap	Exchange one set of cash flows for another at regular intervals	OTC
Option	The *right but not the obligation* to buy (call) or sell (put) by a certain date at a certain price	OTC EXCHANGE

That right without an obligation is the key distinction between an option and a forward or futures contract.

Recap: Derivative

Over-the-counter OTC

- Network of dealers; major banks act as **market makers**, quoting bids and offers
- Contracts are **customised**, governed by master agreements
- Where the $846 trillion lives

Exchange-traded EXCHANGE

- Standardised contracts on organised exchanges
- CME Group, ICE, Eurex, ASX, China Financial Futures Exchange, NSE India (world's busiest by contracts), B3
- Exchange clearing house guarantees performance (Week 2)

A simple comparison?

Recap: Derivative

Market maker in the OTC market

- An OTC contract is made directly between two parties.
- One of the two parties can be a market maker, usually a bank.
- A market maker **regularly quotes** two prices: (1) bid: the price at which it buys; (2) offer: the price at which it sells
- It stands ready to trade and therefore creates an available market.
- **Without a market maker:** Two ordinary participants trade directly only when their need happen to match.

Problem 1.7

- *(a) What is the difference between the over-the-counter and the exchange-traded market? (b) What are the bid and offer of a market-maker in the over-the-counter market?*

Answer 1.7

(a)

- The **OTC market** is a telephone- and computer-linked network of financial institutions, fund managers and corporate treasurers where two participants can enter into any mutually acceptable contract.
- An **ET market** is a market organised by an exchange where traders either meet physically or communicate electronically and the contracts that can be traded have been defined by the exchange.

Answer 1.7

(b)

- When a market maker quotes a *bid* and an *offer*, the **bid** is the price at which the market maker is prepared to buy and the **offer** is the price at which the market maker is prepared to sell.

Recap: Derivative

Stock vs. ET stock option

- Newly issued stock: sold by the company to investors.
- The company receives funds from the stock issue.
- ET stock option: traded between investors on an exchange.
- The company is not directly involved in the option trade.
- Therefore, the option trade **does not** raise funds for the company.
- Key idea: 'first issued' refers to the **primary market**. Later trading of existing shares between investors also does not normally provide new funds to the company (**secondary market**).

Problem 1.9

- *A stock when it is first issued provides funds for a company. Is the same true of an exchange-traded stock option? Discuss.*

Answer 1.9

- **No**
- An exchange-traded stock option provides no funds for the company. It is a security sold by one investor to another. The company is not involved.
- By contrast, a stock when it is first issued is sold by the company to investors and does provide funds for the company.

Recap: Derivative

What are derivatives used for?

- **Hedging:** reduce or remove a price risk you already have by taking an offsetting position in a derivative, thereby locking in certainty;
- **Speculation:** take a view on where the market is going and using derivatives to profit if your view is correct;
- **Arbitrage:** lock in a riskless profit from mispricing by buying at the lower price and selling at the higher price.

First ask whether the participant already has an **exposure**.

Recap: Derivative

Example 1.1 Hedging with forward contracts

It is May 13, 2015. ImportCo must pay £10 million on August 13, 2015, for goods purchased from Britain. Using the quotes in Table 1.1, it buys £10 million in the three-month forward market to lock in an exchange rate of 1.5742 for the pounds it will pay.

ExportCo will receive £30 million on August 13, 2015, from a customer in Britain. Using quotes in Table 1.1, it sells £30 million in the three-month forward market to lock in an exchange rate of 1.5736 for the pounds it will receive.

Table 1.1 Spot and forward quotes for the USD/GBP exchange rate, May 13, 2015 (GBP = British pound; USD = U.S. dollar; quote is number of USD per GBP)

	Bid	*Offer*
Spot	1.5746	1.5750
1-month forward	1.5742	1.5747
3-month forward	1.5736	1.5742
6-month forward	1.5730	1.5736

ImportCo: afraid that GBP will appreciate
>> buying GBP later would cost more USD
ExportCo: afraid that GBP will depreciate
>> future GBP receipts would covert into fewer USD

>> Delivery price (K): From the bank's perspective
Bid: the price at which the bank buys GBP
Offer/Ask: the price at which the bank sells GBP

Source: textbook, p.31

Recap: Derivative

Timeline

- Time 0: May 13, 2025 (Today)
  - ImportCo agrees to buy (**long**) GBP forward at the bank's offer price;
  - ExportCo agrees to sell (**short**) GBP at the bank's bid price.
  - Usually, no cash flow occurs at Time 0.
- Time T: Aug 13, 2025 (In 3 months)
  - ImportCo pays USD to the bank and receives GBP to pay its UK supplier.
  - ExportCo receives GBP from its UK customer, sells GBP to the bank, and receives USD.

Recap: Derivative

Options can also be used for speculation. Suppose that it is October and a speculator considers that a stock is likely to increase in value over the next two months. The stock price is currently \$20, and a two-month call option with a \$22.50 strike price is currently selling for \$1. Table 1.5 illustrates two possible alternatives assuming that the speculator is willing to invest \$2,000. One alternative is to purchase 100 shares.

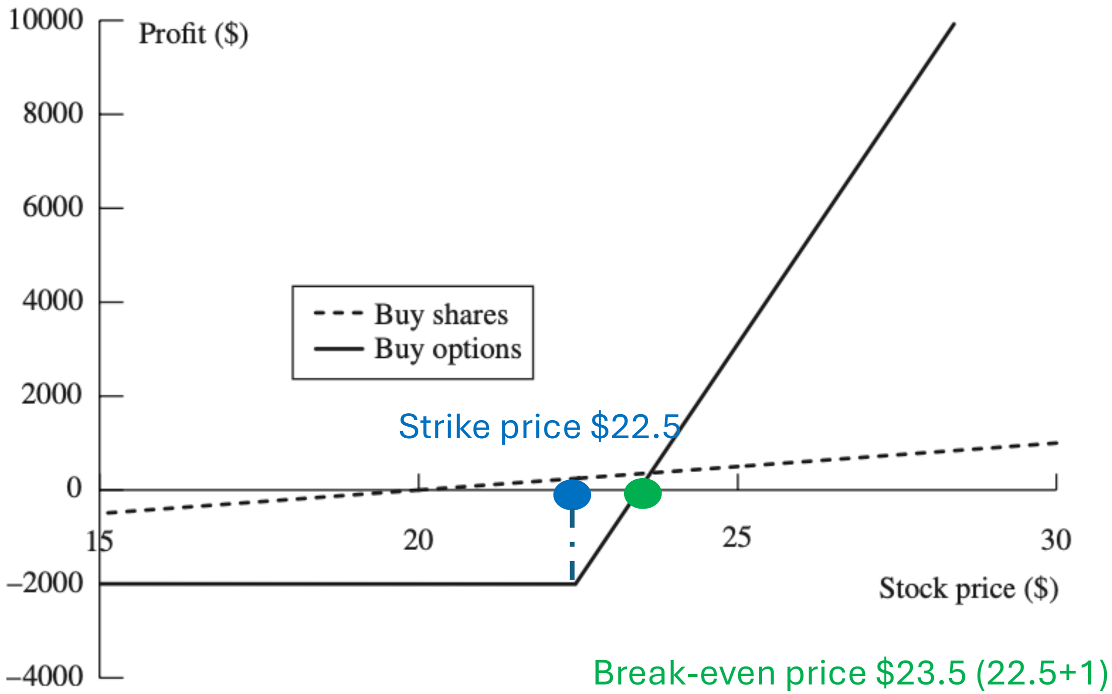
Table 1.5 Comparison of profits from two alternative strategies for using \$2,000 to speculate on a stock worth \$20 in October

Speculator's strategy	*December stock price*	
	\$15	*\$27*
Buy 100 shares	–\$500	\$700
Buy 2,000 call options	–\$2,000	\$7,000

If the stock rises to \$27, buying shares yields a profit of \$700 ((27-20)*100), whereas buying call options yields \$7,000 ((4.5*2000)-2000) because each option has a payoff of \$4.50.
However, if the stock falls to \$15, the share investment loses \$500 ((15-20)*100), while the options expire worthless, resulting in a \$2,000 loss.

>>Good outcomes become very good, while bad outcomes result in the whole initial investment being lost!

Source: textbook, p.33



>>Key idea: Options like futures provide a form of **leverage**. For a given investment, the use of options magnifies the financial consequences.

Recap: Derivative

Example 1.3 An arbitrage opportunity

A stock is traded in both New York and London. The following quotes have been obtained:

New York: \$152 per share
London: £100 per share
Value of £1: \$1.5500

\$155 per share >> the stock is 3 dollars more expensive in London!

A trader does the following:

1. Buys 100 shares in New York
2. Sells the shares in London
3. Converts the sale proceeds from pounds to dollars.

This leads to a profit of

$$100 \times [(\$1.55 \times 100) - \$152] = 300$$

Source: textbook, p.35

Problem 1.2

- *Explain carefully the difference between: (a) hedging, (b) speculation, and (c) arbitrage.*

Answer 1.2

- A company is ***hedging*** when it has an exposure to the price of an asset and takes a position in futures or options markets to **offset the exposure.**
- In a ***speculation*** the company **has no exposure to offset.** It is betting on the future movements in the price of the asset.
- ***Arbitrage*** involves taking a position in two or more different markets to lock in a profit.

Problem 1.18

- *An airline executive has argued: “There is no point in our using oil futures. There is just as much chance that the price of oil in the future will be less than the futures price as there is that it will be greater than this price.” Discuss the executive’s viewpoint.*

Answer 1.18

- It may well be true that there is just as much chance that the price of oil in the future will be above the futures price as that it will be below the futures price.
- This means that the use of a futures contract for **speculation** would be like betting on whether a coin comes up heads or tails.
- But it might make sense for the airline to use futures for **hedging** rather than speculation.

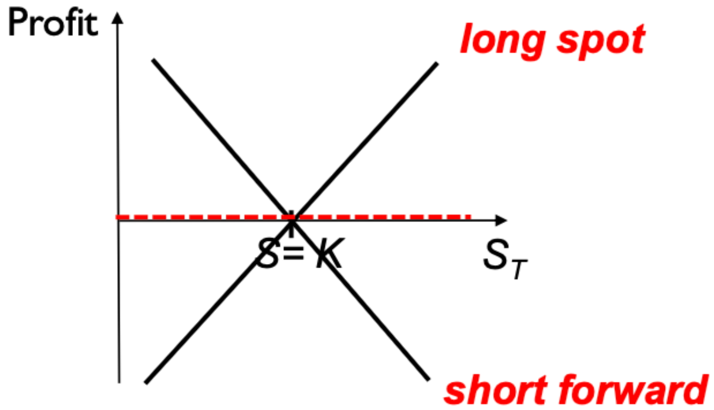
Answer 1.18

- The futures contract then has the effect of reducing risks.
- It can be argued that an airline should not expose its shareholders to risks associated with **the future price of oil** when there are contracts available to hedge the risks.

Recap: Forward-contract payoff

Profit on short forward = $-(S_T - K)$
Profit on long spot = $(S_T - \cancel{S})$ K
Overall profit = 0 (if S = K)

K = delivery price under forward contract
S_T = spot price of underlying asset at maturity



Problem 1.4

- *An investor enters into a short forward contract to sell 100,000 British pounds for Australian dollars, at an exchange rate of 1.6400 Australian dollars per pound. How much does the investor gain or lose if the exchange rate at the end of the contract is: (a) 1.6000, (b) 1.7000?*

Answer 1.4

Profit on short forward = $-(S_T - K)$

(a)

- The investor is obligated to sell 100,000 pounds for \$1.6400 (K) when they are worth \$1.6000 (S_T).
- The gain is 100,000 x \$0.04 (-(1.6-1.64)) = \$4,000.

(b)

- The investor is obligated to sell 100,000 pounds for \$1.6400 (K) when they are worth \$1.7000 (S_T).
- The gain is 100,000 x -\$0.06 (-(1.7-1.64)) = -\$6,000 (i.e. loss).

Problem 1.20

- *A trader enters into a short forward contract on 100 million yen. The forward exchange rate is $0.0080 per yen. How much does the trader gain or lose on the forward contract if the exchange rate at the end of the contract is (a) $0.0074 per yen; (b) $0.0091 per yen?*

Answer 1.20

$$\textbf{Profit on short forward} = -(S_T - K)$$

(a)

- The trader sells 100 million yen for \$0.0080 per yen (K) when the exchange rate is \$0.0074 per yen (S_T). The gain is 100 x 0.0006 (-(0.0074-0.0080)) millions of dollars or \$60,000.

(b)

- The trader sells 100 million yen for \$0.0080 per yen (K) when the exchange rate is \$0.0091 per yen (S_T). The loss is 100 x 0.0011 (-(0.0091-0.0080)) millions of dollars or \$110,000.